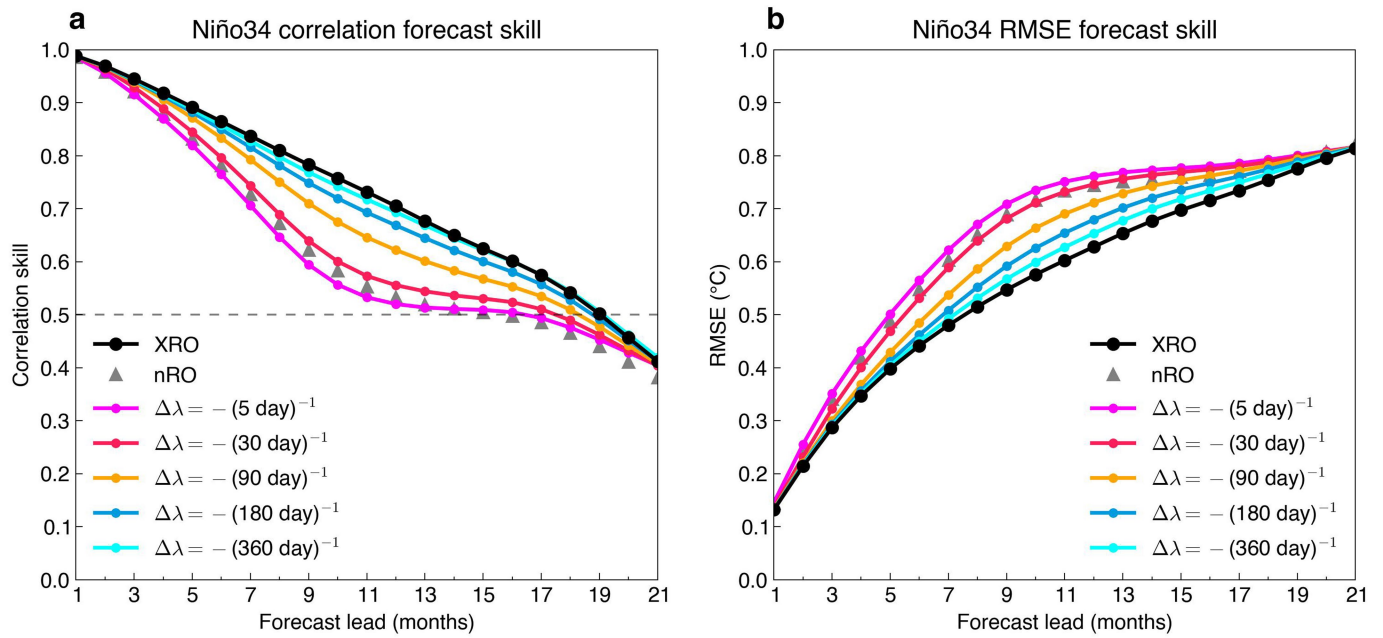


Influence of the "memory" effect of other climate modes on ENSO forecast skill



Extended Data Fig. 4 | Influence of the memory effect outside the equatorial Pacific on ENSO forecast skill. Shown are the all-months correlation skill (a) and RMSE (b) of the 3-month running mean Niño3.4 index, as a function of the forecast lead month in the XRO forecast (black), the nRO forecast (grey triangle), and the "Losing memory" sensitivity experiments

(colour curves) by adding different damping rates (ranging from a strong damping rate of $-(5 \text{ day})^{-1}$ implying no memory to a weak damping rate of $-(360 \text{ day})^{-1}$ implying longer memory) to the non-ENSO modes (See "Losing memory experiments" in Methods). The initial condition memory effect of the climate modes outside equatorial Pacific extends the skill of ENSO forecasts.